

Data Collection & Preparation for Green Jobs Analysis

Week 2 Deliverable — Data Collection Strategy Document

Junior Data Analyst — Green Jobs & Sustainability Internship

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1. Objective

The Week 1 strategy document identified ten candidate public data sources spanning workforce, energy, and environmental data. This Week 2 deliverable narrows that list to three primary sources, documents the criteria used to select them, and lays out a reproducible, step-by-step methodology for extracting, cleaning, and preparing the data so it is ready for exploratory analysis in Week 3. The guiding principle throughout is that every decision — which source to use, which rows to drop, how a missing value is treated — should be explainable and repeatable by another analyst, not a one-off judgment call.

2. Selection Criteria for Data Sources

Out of the ten sources identified in Week 1, three were selected as primary datasets for this phase. The selection was governed by five criteria, applied consistently across all candidates:

Criterion	What It Means for This Project
Credibility	Source is a government body, UN/multilateral agency, or an established research institute — not a blog, forum, or unverified aggregator.
Currency	Dataset has been updated within the last 12–18 months, or the underlying survey/questionnaire cycle is annual, so trend analysis in Weeks 3–4 is not built on stale figures.
Licensing / Openness	Data is explicitly published for public or academic reuse, with no paywall or restrictive terms that would block redistribution in this report.
Granularity	Data is available at country level at minimum, and at state/UT level where possible (important for the India-specific research questions from Week 1).
Relevance to KPIs	Directly maps to at least one of the seven KPIs defined in the Week 1 strategy — sources that don't were excluded even if credible, to keep the scope tight.

Sources such as Kaggle/Google Dataset Search and Our World in Data remain useful as secondary, cross-check references (Section 6), but were not selected as primary sources because they re-package data already captured by IRENASTAT, ILOSTAT, or government portals — using them as primary sources would risk double-counting the same underlying figures.

3. Selected Data Sources & Extraction Method

Source	Why It Was Selected	Extraction Method
IRENASTAT (IRENA Renewable Capacity Statistics)	Only global, standardized, annually-updated dataset on installed renewable capacity by country and technology. Directly feeds the “Installed renewable capacity (MW)” KPI.	IRENASTAT online query tool (pxweb.irena.org) — filter by country/technology/year, export as CSV. No account or API key required.
ILOSTAT — Employment by Economic Activity (ISIC)	Authoritative, internationally comparable employment data. Sectors aligned to ISIC codes can be mapped to green-relevant industries (renewable	ILOSTAT bulk data download (CSV via ilostat.ilo.org) or ILOSTAT API (REST, JSON/CSV response) filtered by country and indicator code.

Source	Why It Was Selected	Extraction Method
	energy, waste management, forestry). Feeds workforce KPIs.	
NITI Aayog SDG India Index & Dashboard	Only consolidated, state-wise sustainability performance score for India, aligned to SDG 7, 8, 12, and 13 — directly supports the geographic/state-level comparison planned in the Week 1 strategy.	Published as an Excel/PDF report and an interactive dashboard (sdgindiaindex.niti.gov.in); index scores extracted manually into CSV since no bulk API is offered.

All three extractions are done through each provider's own free export tool (CSV or Excel) rather than screen-scraping, which keeps the pipeline compliant with each source's terms of use and easy for another analyst to reproduce.

4. Data Cleaning & Preparation — Step-by-Step Guide

The following seven-step process is applied to each of the three datasets independently before they are merged into one analysis-ready table.

Step 1 — Initial Inspection

For each raw export, before any changes are made, capture a baseline: row/column counts, data types per column, and the percentage of missing values per column. This baseline is saved alongside the cleaned file so any data loss in later steps is traceable.

```
import pandas as pd
```

```
df = pd.read_csv('irena_capacity_raw.csv')
print(df.shape)
print(df.dtypes)
print(df.isna().mean().sort_values(ascending=False))
```

Step 2 — Standardize Column Names & Structure

Rename columns to a consistent snake_case schema (e.g., country, year, technology, capacity_mw) across all three datasets, since IRENA, ILO, and NITI Aayog each use different header conventions. This is what makes the later merge (Step 6) possible without manual re-mapping every time.

Step 3 — Standardize Country / Region Names

Country and state names are the most common source of silent join failures — e.g., “India” vs. “IN” vs. “Republic of India”. Map every dataset's location field to ISO-3166 country codes (and, for the NITI Aayog dataset, a fixed list of the 28 states + 8 union territories) using a lookup table, so downstream joins match on a single reliable key.

Step 4 — Handle Missing Values

- Structural gaps (a country simply has no renewable capacity of a given technology): fill with 0, not blank — the absence is a real value, not missing data.

- Reporting gaps (a country skipped a survey year): leave as null and flag with a boolean `*is_imputed*` column rather than silently interpolating, so later analysis can choose whether to include or exclude them.
- Where an entire year is missing for an otherwise consistent series, use linear interpolation only, and only across gaps of one year — never extrapolate beyond the observed range.

Step 5 — Standardize Units & Formats

- Capacity figures: confirm all values are in megawatts (MW); IRENA occasionally reports select technologies in GW in summary tables.
- Dates: convert every year field to a plain integer (e.g., 2024), removing any embedded fiscal-year notation (“FY2023-24”) from the NITI Aayog export.
- Currency (where investment figures are used later): standardize to USD using a documented, fixed conversion-rate source rather than mixing INR and USD figures in the same column.

Step 6 — De-duplicate and Merge

After each dataset is individually cleaned, de-duplicate on the natural key (country/state + year + category) and merge the three tables into one long-format analysis table keyed on country/state and year. India-specific rows are used for the state-level SDG comparison; the full country list is retained for the global capacity-vs-employment correlation analysis.

```
merged = (
    capacity_clean
    .merge(employment_clean, on=['country_iso3', 'year'], how='left')
    .merge(sdg_india_clean, on=['state', 'year'], how='left')
)
merged.to_csv('green_jobs_analysis_dataset.csv', index=False)
```

Step 7 — Validate & Document

- Re-run the Step 1 inspection on the cleaned/merged output and compare row counts and null rates against the baseline to confirm no unintended data loss.
- Spot-check 5–10 known values (e.g., India's 2023 installed solar capacity) against the original source publication to confirm the pipeline preserved the correct figures.
- Record every transformation decision (fills, interpolations, unit conversions) in a short data dictionary saved alongside the cleaned CSV, so Week 3's exploratory analysis starts from a fully documented dataset.

5. Recommended Tools & Techniques

- Python (pandas) for all cleaning and merging steps — free, scriptable, and reproducible end-to-end.
- OpenRefine for one-off fuzzy-matching of inconsistent country/state name spellings before they're finalized in the lookup table.
- Google Sheets for the NITI Aayog data, since it currently must be extracted from published PDF/Excel reports rather than an API, before importing into the Python pipeline.
- Jupyter Notebook to keep the cleaning steps, code, and inline notes together as one auditable record.

6. Anticipated Data-Quality Challenges

- Mismatched reporting years — IRENA reports calendar-year data while NITI Aayog's index sometimes aligns to Indian fiscal years; this is resolved by explicitly documenting the year-mapping rule used in Step 5.
- Sector classification differences — ILOSTAT's ISIC codes do not map one-to-one onto “green jobs”; a documented mapping table (built by cross-referencing ILO's own green-jobs classification guidance) is maintained and included with the cleaned data so the mapping logic is transparent, not implicit.
- India state-level granularity — IRENA and ILOSTAT report at country level only; where state-level detail is needed, data.gov.in's MNRE state-wise capacity figures are used as a secondary cross-check source (Section 3) rather than left as a gap.
- No proprietary access — confirmed only free, publicly exportable data was used across all three sources; no paid API tiers or subscription datasets are required for this pipeline.

7. Conclusion

This document narrows the Week 1 source list to three defensible primary datasets, explains why each was chosen, and lays out a seven-step cleaning process that turns three structurally different public exports into one merged, analysis-ready table. The resulting dataset — with its accompanying data dictionary — is the direct input for Week 3's exploratory data analysis, so the quality decisions made here (how missing values are treated, how countries and states are matched) will shape everything that follows.